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A new quantitative Raman measurement scheme using Teflon as a novel intensity correction standard as well as the sample container

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A novel and reliable quantitative Raman measurement scheme has been proposed for the analyses of ethanol and isopropanol solutions. Teflon tubing was employed as an effective intensity correction standard as well as the sample cell. This allowed for the synchronous collection of the mixed Raman spectrum from the Teflon standard and the sample without any extra optical configuration. A non-overlapping Teflon band was used as the standard peak to correct the Raman intensity changes resulting from laser power variations. In addition to the use of Teflon tubing,a wide area illumination (WAI) scheme was employed, which made the laser illumination into a circle of 6 mm diameter (area: 28.7 mm^2), to cover a wide sample area. The WAI scheme improved the reliability of the Raman measurement by significantly increasing the reproducibility of the sampling owing to a decreased sensitivity to sample placement with respect to the focal plane as well as a wider sample coverage area. The resulting Raman spectra were more reproducible and more representative of the correct sample composition. Overall, a superior prediction performance was achieved using the WAI scheme compared to the conventional Raman collection method. The proposed system has great potential for use in the quantitative analysis of diverse liquid samples with good reproducibility. Copyright © 2006 John Wiley & Sons, Ltd.

KEYWORDS: Raman spectroscopy; Teflon; synchronous standard; on-line analysis

INTRODUCTION

Raman spectroscopy provides selective and diverse information regarding a sample's molecular structure and chemical composition. The Raman instrumentation scheme is relatively simple, is capable of measuring diverse samples without pre-treatment and can be easily modified for on-line analysis by using an optical fibre. Additionally, because the Raman signal is not sensitive to the presence of water, it can be effectively utilized for analyses in aqueous environments. In spite of its distinct advantages, Raman spectroscopy has not been exploited for quantitative analysis, unlike other competing methods such as near-infrared (NIR) and IR spectroscopy, because the Raman intensity is relatively less reproducible owing to factors such as variations of laser power, sample presentation and optical arrangement. Therefore, it is necessary to have an internal or external standard that can correct the changes in the Raman spectrum to achieve a reliable quantitative analysis. Additionally, it has low sensitivity when it is not based on surface-enhanced Raman scattering (SERS).

The correction of the Raman signal using an external standard requires the continuous collection of separate Raman spectra for the corresponding standards and samples.[1–5] When an external standard is used, the collection of the Raman spectra of the sample and standard are asynchronous and, therefore, the Raman spectrum of the standard may not perfectly represent the variations in the spectrum of the sample.

An internal standard is a material that is incorporated into the sample. The resulting Raman feature is a mixture of that of the sample and the internal standard, and so non-overlapping bands of the internal standard can be used to correct for any Raman intensity variations of the sample.[6–8] The use of an internal standard can be more appropriate for the intensity corrections since the Raman spectral collection is synchronous for both the sample and standard in the same matrix. However, it is cumbersome in routine analysis because of the need for the addition of the internal standard at each measurement. When an automated

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fluidic system using Raman spectroscopy is considered for high-throughput analysis, the addition of an internal standard is impossible.

There is a strong demand for a new Raman method that can overcome the drawbacks of the existing external and internal standards methods and simultaneously be suitable for continuous routine or on-line measurements. Two research groups have recently utilized external standard materials such as diamond[9] and sapphire[10] positioned in front of the sample (ethanol solution). The laser illumination is through these standards to collect the Raman spectra. However, these approaches require an extra optical configuration that must be precisely aligned at all times to give a reproducible Raman signal for the standard. A simpler, more cost-effective and more robust spectral collection strategy is highly desirable.

We have proposed a novel Raman collection scheme using Teflon (perfluoroalkoxy fluorocarbon, PFA) tubing as an effective signal correction standard and sample container (cell) simultaneously without the requirement of any extra optical configuration. The sample is transferred into a Teflon tubing and the Raman spectrum of the sample is collected directly through the Teflon tubing. The resulting spectrum is the sum of spectral features from the sample and the Teflon tubing, which allows the use of non-overlapping Teflon bands to correct for variations in the Raman intensity. Furthermore, Teflon is chemically inert, resists even strong acids, is thermally stable up to 260 °C and is flexible, which allow easy manipulation for on-line measurements at minimal cost.

To improve the reproducibility of direct Raman spectroscopy measurements through the Teflon tubing, a wide area illumination (WAI) scheme was used, which used a circular laser beam of 6 mm diameter (area: 28.7 mm^2) and a focal length of 250 mm, as shown in Fig. 1. By illuminating a large surface area, the resulting Raman spectrum can give more accurate quantitative information and the signal of the standard (Teflon) can be more reproducible because of its lower sensitivity to sample placement with respect to the focal plane.

To evaluate the validity the proposed scheme, two different chemical matrices consisting of ethanol/water and isopropanol/water were employed for quantitative analyses. Ethanol is one of the most popular organic solvents in the pharmaceutical industry. Isopropanol is frequently used for the cleaning of wafers in the final step of wet processes,[11] and the accurate determination of the isopropanol concentration is critical in order to maximize the utilization of the drying equipment. Since Teflon tubing is extensively used in wet stations, the ability to make direct Raman measurements without any process system modifications, as shown in Fig. 1, should be highly beneficial and advantageous. For calibration, both the methods of integrated area and principal component regression (PCR)[12–15] were used.

EXPERIMENTAL

Sample preparation

Three sets of ten different ethanol solutions (concentration range: 0.23–8.97 wt%) were prepared by mixing the appropriate quantity of ethanol in purified water. The first set was used for the development of the calibration model and the other two sets were used for the validation of the prediction performance. The concentrations of each dataset were randomly selected within a given range.

Four sets of ten different isopropanol solutions (concentration range: 0.07–9.89 wt%) were prepared. The concentration range was designed to comply with those employed in the electronics industry. The concentrations of each dataset were randomly selected within a given range. The Raman spectra of each set were collected over several days to include possible instrumental variation or drift.

Wide area illumination (WAI) scheme

A schematic diagram of the WAI scheme (PhAT System Raman Analyser, Kasier Optical Inc., Ann Arbor, MI, USA) is shown in Fig. 1. The laser beam (785 nm, diode laser) is

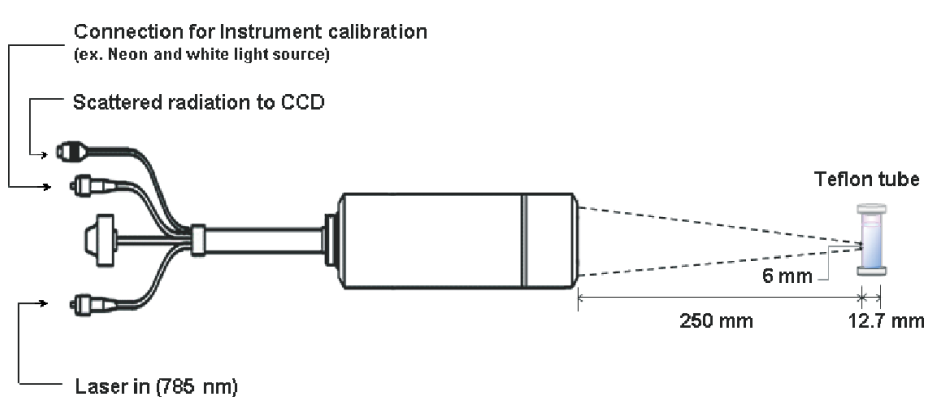

Figure 1. Schematic diagram of the wide area illumination (WAI) scheme that uses a circular laser beam of 6 mm diameter (area: 28.7 mm^2) and a focal length of 250 mm. The Teflon tube is positioned at the focal point. This figure is available in colour online at www.interscience.wiley.com/journal/jrs.

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introduced by ten individual optical fibres simultaneously and each optical fibre illuminates a slightly different spot to finally form a circular illumination area of 6 mm diameter (area: 28.7 mm^2) to cover a large sample area. Conventional dispersive and FT-Raman spectroscopic systems have an illumination spot ranging in size from 100 to 500 μm; therefore, the surface coverage area of the WAI scheme is 140–200 times larger than the typical area. The scattered radiation is then collected by each optical fibre and delivered to the charge-coupled device (CCD) detector. The dispersed radiations via a holographic transmission grating from each optical fibre were separately collected in different rows of the CCD and the final Raman spectrum was generated by integrating the Raman signal delivered from all ten optical fibres. An extra optical fibre is used for connection to the neon and white light sources when wavelength and intensity calibrations are necessary.

The optimized focal length of the WAI was 250 mm and the Teflon tube was located at the focal point. A readily available, half-inch Teflon tube (PFA fluorocarbon, outer diameter: 12.7 mm, inner diameter: 9.68 mm) was used. PFA is fairly transparent in the visible and NIR ranges as shown in a previous publication.[16] The optical quality of Teflon is not necessarily as consistent as quartz or CaF_2; however, it has been shown that the optical quality as well as thickness of a Teflon tube is reproducible for optical measurements.[16]

In addition to the WAI method, a conventional Raman scheme with 10 mm focal length was employed. In this case, the Teflon tube was located in front of the Raman probe and its position was adjusted to locate the focal point at the centre of the Teflon tube. In both schemes, the distance between the Teflon tubing and the Raman probe was fixed throughout the whole experiment to minimize the possibility of fluctuation in the Raman signal due to variations in distance. Two separate dispersive Raman spectrometers equipped with a diode laser (785 nm), a CCD detector and a holographic grating were used for each scheme (Kasier Optical Inc., Ann Arbor, MI, USA). Each Raman spectrum was collected with an exposure time of 10 s (1 scan). All spectral processing including baseline correction, intensity correction and PCR were accomplished using Matlab version 7.0 (The MathWorks Inc., MA, USA).

RESULTS AND DISCUSSION

Raman spectral features of Teflon and ethanol solution

The spectra of the empty Teflon tube (PFA fluorocarbon) and ethanol in the Teflon tube are shown in Fig. 2. The molecular structure of PFA is also shown in Fig. 2. The presented spectrum was collected by the WAI scheme. There are seven Teflon bands in the 1600–200 cm^{-1} range. The 732 cm^{-1} band shows the highest intensity and corresponds to the stretching mode with A_1 symmetry of CF_4. The two bands at 384 and 291 cm^{-1} originate from the splitting of the

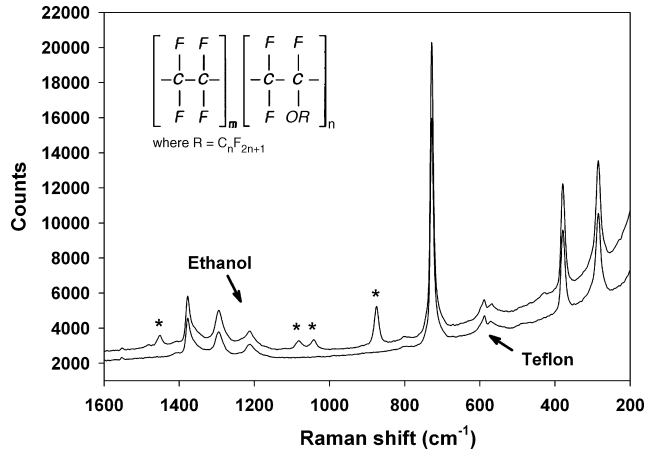

Figure 2. Raman spectra of an empty Teflon tube (PFA: perfluoroalkoxy fluorocarbon) and 99.9% ethanol collected in the Teflon tube. The molecular structure of PFA is also shown.

E symmetry mode. The three small bands in the 1400–1150 and 600–530 cm^{-1} ranges result from the splitting of two F_2 symmetry lines.[17]

If at least one non-overlapping Teflon peak out of the seven peaks is present during the actual collection of the Raman spectra of the sample, this peak can be used to compensate for variations in Raman intensity in a synchronous fashion. The Raman peaks of ethanol (99.9%) collected using the Teflon tubing are marked with an asterisk in the figure. Without question, the resulting spectrum is the sum of the ethanol and Teflon spectra. The spectral features of ethanol are simple, and the major band is centred at 875 cm^{-1}, which does not overlap with the bands from Teflon. Two other smaller bands were observed at 1452 cm^{-1} and in the 1100–1011 cm^{-1} range. The 930–685 cm^{-1} range that included the ethanol band at 875 cm^{-1} was solely used for calibration since its intensity was the strongest and the strong Teflon peak nearby at 732 cm^{-1} could be used as the standard peak for the Raman intensity correction.

Figure 3 shows the Raman spectra (930–685 cm^{-1}) for four different ethanol concentrations before and after the intensity correction. The results from both Raman collection schemes (conventional and WAI) are also shown in the figure. The relative intensity of Teflon to ethanol is much higher for the WAI scheme since Teflon is much denser (solid) and stronger in Raman scattering. Therefore when a wider area is illuminated, the more dominant spectral feature of Teflon can be observed. Before intensity correction, minor baseline variations were present in the sample spectra. To correct the Raman intensity, the baselines of the spectra were initially corrected at two points (930 and 685 cm^{-1}) and zeroed, and the area of the Teflon band in the 780–685 cm^{-1} range was calculated. Finally, each sample spectrum was divided by the corresponding area of the Teflon spectrum over the 780–685 cm^{-1} range. In this way, unwanted intensity variations not related to variations in concentration could be

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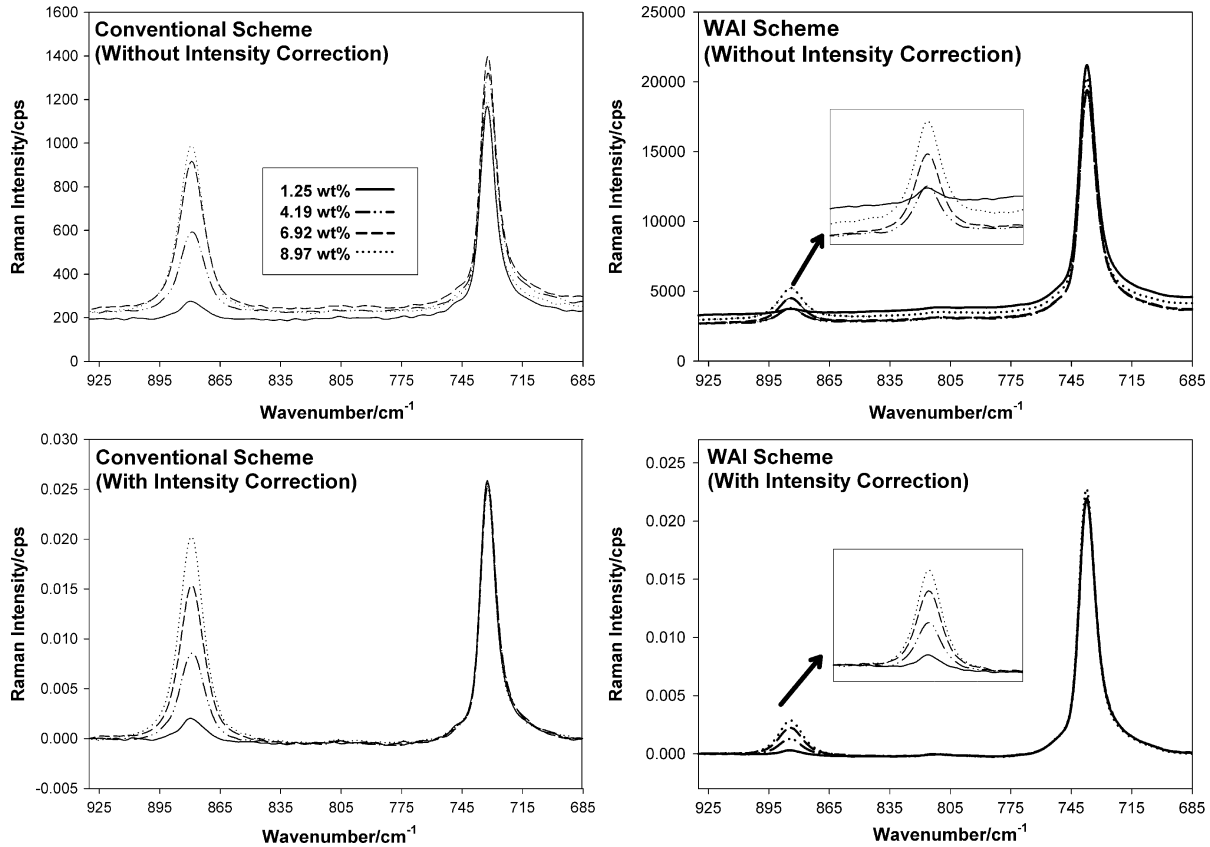

Figure 3. Raman spectra (930–685 cm^{-1} range) of four different ethanol concentrations before and after intensity correction. The spectra were collected using both Raman collection schemes (the conventional and WAI).

effectively corrected. With the use of the intensity correction based on the Teflon peak, the Raman intensity varied according to the changes in the ethanol concentration and no significant differences were observed between both the Raman collection schemes by simple visual inspection.

Quantitative calibration for the determination of ethanol

Since the ethanol peak is distinct without spectral overlapping, the simple integrated area method as well as the PCR was simultaneously utilized to determine the ethanol concentration. The calibration models were developed using the first dataset (ten samples) and its prediction performance was evaluated by predicting the spectra (20 spectra) in two other datasets (validation set). As mentioned, the spectra in each dataset were collected at separate times. For the integrated area method, only the intensity-corrected Raman spectra using the Teflon peak were used. Both Raman spectra, with and without the intensity correction using the Teflon peak, were used for PCR. Simultaneously, the overall results obtained by using the conventional and WAI schemes were compared. The 921–8301 cm^{-1} and 930–685 cm^{-1} ranges were used for integrated area calculation and PCR, respectively.

The overall results are summarized in Table 1. The numbers in parentheses correspond to the number of factors used. As shown, PCR (multivariate calibration method) provided better calibration results for both cases compared to the univariate method of integrated area. Figure 4 shows the concentration correlation plots using PCR that resulted from the use of the conventional and WAI schemes. Open and filled circles correspond to the calibration and prediction data, respectively. Without question, the standard error of prediction (SEP) was superior when the Raman spectra were corrected using the Teflon band in both cases.

Along with the intensity correction, the use of the WAI scheme leads to a superior prediction performance over the conventional scheme. Since the Teflon tubing was removed to add a new sample and then re-positioned, it was possible that the position of the Teflon tube could slightly vary from measurement to measurement, even though its positioning was tightly controlled. This could introduce minute variations in the distance between the Raman probe (objective lens) and the Teflon tubing. Additionally, the spots illuminated on the Teflon tube may not be fixed throughout the experiment. This eventually may result in a variation of the intensity ratio between the Teflon and ethanol signals, and adversely affect the correction of the

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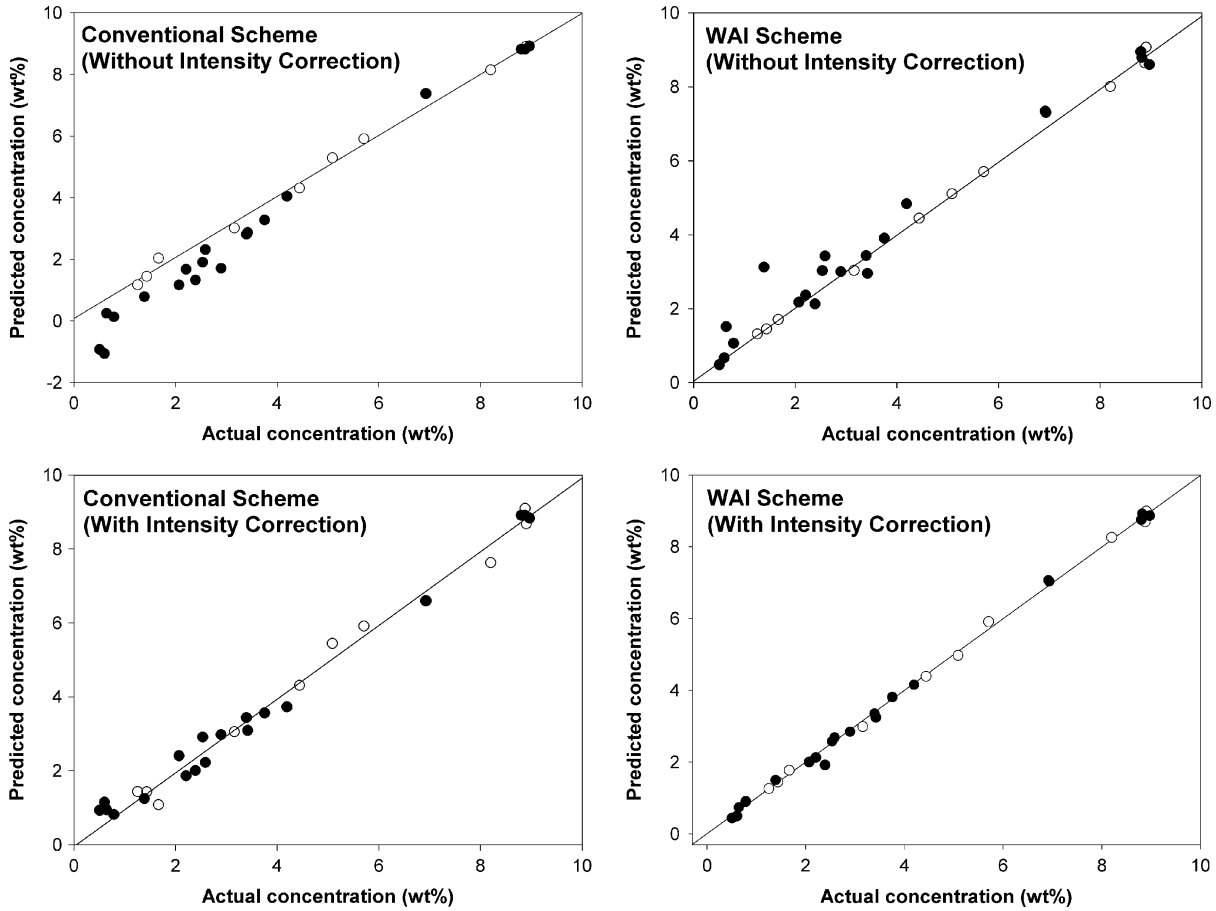

Figure 4. Concentration correlation plots for ethanol that resulted from the use of the conventional and WAI schemes with and without intensity correction. The open and filled circles correspond to the calibration and prediction data, respectively.

Table 1. Integrated area and PCR calibration results for ethanol. The results from the use of the conventional and WAI schemes are compared. The numbers in parentheses correspond to the number of factors used

Calibration method	Scheme	Without intensity correction		With intensity correction	
		SEC[a]	SEP	SEC[a]	SEP
Integrated area	Conventional	–	–	0.26	0.97
	WAI	–	–	0.28	0.37
PCR	Conventional	0.14 (2)	0.88	0.36 (1)	0.60
	WAI	0.25 (2)	0.37	0.10 (1)	0.15

[a] Standard error of calibration.

Raman intensity. The intensity ratio between signals at a given ethanol concentration and the Teflon signal should be constant to be used for an accurate intensity correction protocol. The conventional scheme applied the laser beam over a small area (approximately 80∼100 μm in diameter) and had a focal length that was much shorter than in the WAI scheme. Because of this, the intensity ratio between ethanol and Teflon signals was much more sensitive to slight variations in the distance between the Raman probe and Teflon tube, as well as to different laser illumination spots on the Teflon tube. This phenomenon can be observed in Fig. 5, which shows ten Raman spectra (without intensity correction) of 8.97% ethanol collected by the conventional (top) and WAI (bottom) schemes. The Teflon tube was re-positioned for each measurement. As shown, a larger spectral variation was observed when the conventional scheme was used, even though the same ethanol concentration was measured by both methods. More reproducible spectra were collected using the WAI scheme. For quantitative comparisons, the peak ratio between the Teflon peak at 732 cm^{-1} and the ethanol peak at 875 cm^{-1} was calculated. Then, the corresponding average, standard deviation and relative standard deviation (RSD) were calculated. The average and standard deviation of the peak ratio for the conventional scheme were 1.20 and 0.20, respectively, while they were 5.06 and 0.05 for the WAI scheme. The peak ratio between the Teflon and ethanol should be constant in order to be used for correcting the Raman intensity using Teflon as a standard and sample container. The RSD of the peak ratio

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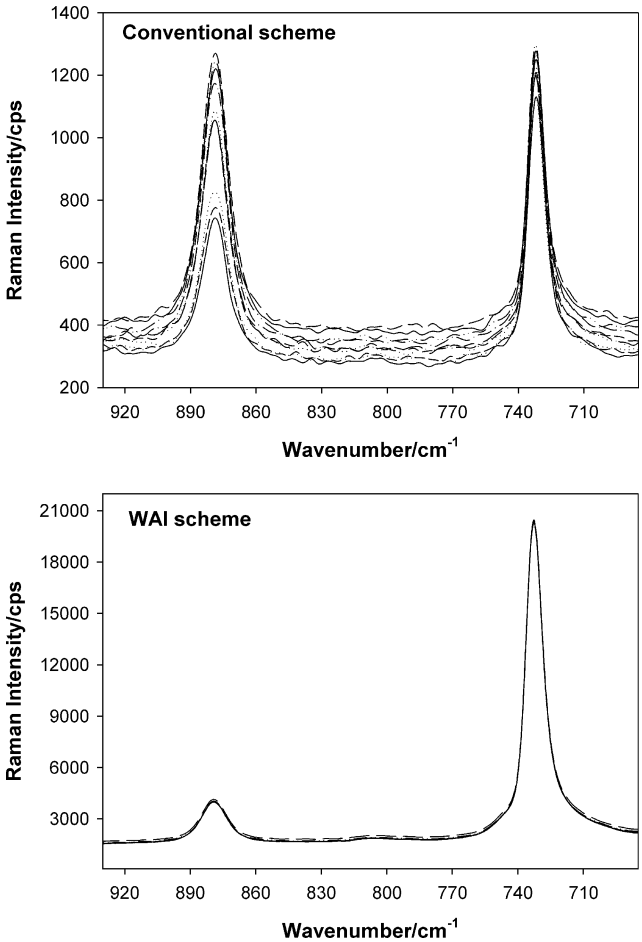

Figure 5. Ten Raman spectra (no intensity correction) of 8.97% ethanol collected by the conventional and WAI schemes.

using the WAI scheme was only 0.9% while it was 16.9% for the conventional scheme. Overall, the WAI scheme provided more quantitatively correct Raman spectral features since it illuminated a much larger area on Teflon to collect the Raman spectra and had a configuration that was less sensitive to sample placement with respect to the focal plane.

Spectral features of isopropanol and range optimization for calibration

Since the WAI scheme was superior in terms of the quantitative spectral representation, Raman spectra of isopropanol solutions were collected with the use of the WAI scheme only. Figure 6 shows the Raman spectra of isopropanol (99.5%) and the empty Teflon tubing. There are six major bands of isopropanol at 1456, 1185, 1093, 946, 816 and 488 cm^{-1} (marked

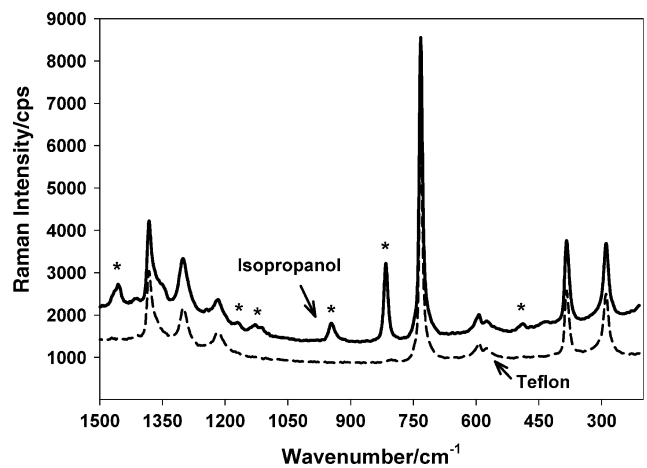

Figure 6. The Raman spectra of an empty Teflon tube and of 99.5% isopropanol collected in the Teflon tube.

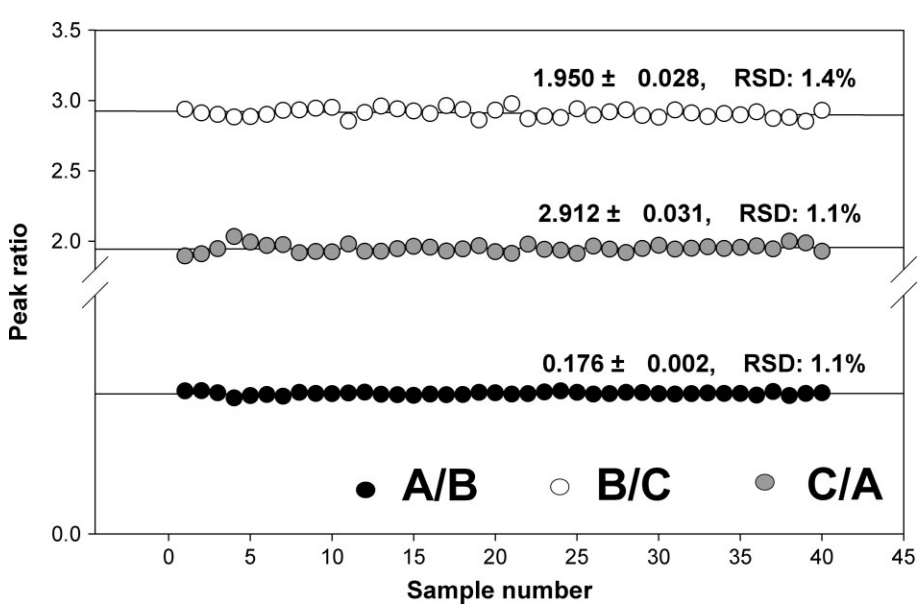

Figure 7. Calculation results of the peak ratios using three Teflon peaks (1382 cm^{-1}, 732 cm^{-1}, 384 cm^{-1} and designated as A, B, C, respectively) that were obtained from 40 independent Teflon spectra.

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with asterisks), and the band at 816 cm^{-1} is the most intense. Since the bands of isopropanol were more spread out over the entire range than those of ethanol, PCR calibration using a wide spectral range might be beneficial since all the information can be utilized. However, this might not be always possible since localized spectral ranges also could provide satisfactory prediction results. It is possible that an optimum spectral range exists that can yield superior multivariate calibration results as long as the Raman intensity is properly corrected. Several chemometric strategies such as moving window PLS (MW-PLS)[18,19] to find the best calibration range have been reported. The main concept of MW-PLS is to perform PLS for all possible combinations of spectral ranges and find the spectral range that yields the minimum error. The same concept can be extended to PCR, the so-called MW-PCR.

To utilize the full spectral range, it is necessary to correct the Raman intensity over the entire region. When the Teflon peak at 732 cm^{-1} is used as a standard to correct a nearby peak such as the band at 816 cm^{-1}, the proportion of increase or decrease between the isopropanol (816 cm^{-1}) and Teflon (732 cm^{-1}) peaks might be the same since these two peaks are located near each other. If one Teflon band is intended to correct an analyte band that is located far away, the proportion of the increase or decrease between both peaks might be slightly different since the quantum efficiency of the CCD pixels could vary in the different spectral ranges. To address this issue, we collected 40 Raman spectra of the empty Teflon tubing by re-positioning the tube between each measurement and selected three Teflon peaks located far away from one another (1382 cm^{-1}, 732 cm^{-1}, 384 cm^{-1} and designated as A, B, C, respectively). Three peak height ratios were calculated (peak height A/peak height B, peak height A/peak height C and peak height B/peak height C). If the peak ratios were constant over 40 measurements without any drift, it could be assumed that the proportion of the increase or decrease of the Raman intensity is constant throughout the entire spectral range. Figure 7 shows the results of the peak ratios calculated from the three Teflon peaks obtained from 40 independent Teflon spectra. As shown, the peak ratios were consistent without any noticeable trend. The corresponding average, standard deviation, and RSD are also summarized in the figure. The average RSD was only 1.2%. Consequently, as long as one non-overlapping Teflon peak was present, it could be used as a standard for the entire spectral range. The 732 cm^{-1} band was used to correct the Raman intensity variation over the whole spectral range.

To perform MW-PCR, the first and second datasets were used as the calibration set and the monitoring set, respectively. At each spectral range, a PCR was performed using the calibration set and the resulting prediction error from the monitoring set was evaluated to determine the best spectral range. The final SEP was calculated by predicting samples in the validation set (third and fourth datasets) using a PCR model developed within the best spectral range

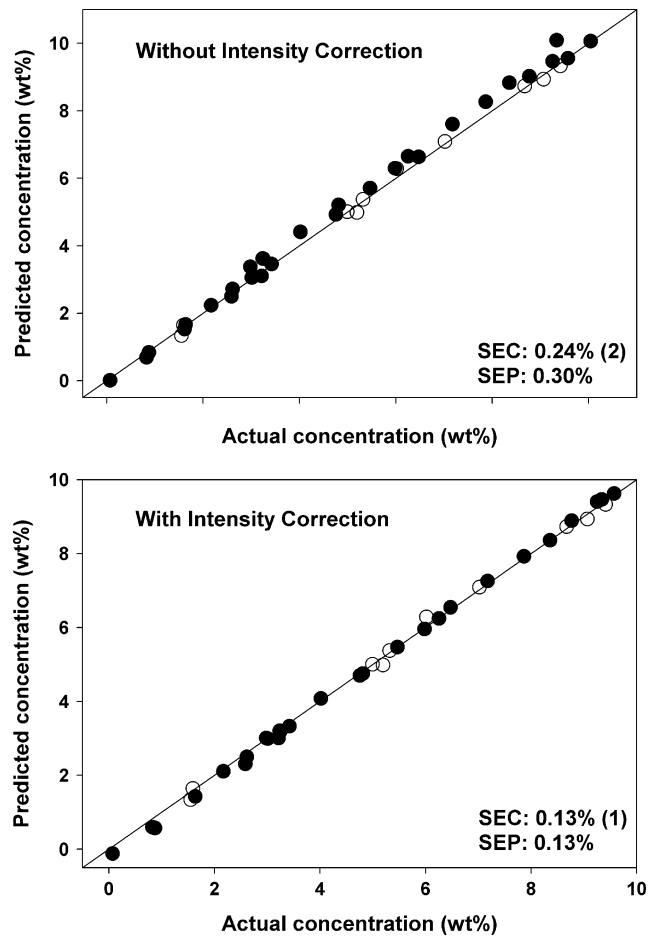

Figure 8. Concentration correlation plots for isopropanol that resulted from the use of the WAI scheme with and without intensity correction. The open and filled circles correspond to the calibration and prediction data, respectively. The PCR calibration results are also summarized in the figure. The numbers in parentheses correspond to the number of factors used.

that was previously determined. To find the best spectral range, the window size (number of spectral points) was varied from 10 to 1300 cm^{-1} with an incremental interval of 2 cm^{-1}. In a given window size, the window was moved from low to high wavenumbers with a 10 cm^{-1} step interval to cover the whole spectral range (1500–200 cm^{-1}). A total of 56 511 spectral ranges with different window sizes as well as different spectral ranges were tested.

The 945–780 cm^{-1} range was selected as the best spectral range. This range incorporated two isopropanol bands centred at 946 and 816 cm^{-1}. The corresponding calibration results are summarized in Fig. 8. The numbers in parentheses correspond to the number of factors used. Figure 8 also shows the concentration correlation plots without and with the intensity correction for the 945–780 cm^{-1} range. The open and filled circles correspond to the calibration and prediction data, respectively. Only PCR was performed to determine

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isopropanol concentration since it showed better calibration performance in the determination of ethanol concentration. Obviously, the intensity correction using the Teflon peak is necessary for a reliable prediction performance. With an intensity correction, the concentrations of isopropanol were accurately predicted with an SEP of 0.13%.

CONCLUSIONS

Direct Raman measurements through a Teflon tubing using a WAI scheme demonstrated reproducible and reliable prediction performance and could be employed for the analyses of diverse liquid samples. The proposed scheme enabled quantitative Raman measurements to be performed simply and accurately without requiring any additional optical configurations for an external standard. At the same time, the WAI scheme successfully generated reproducible Raman spectra that could be considered for practical Raman applications. If a similar WAI scheme with longer focal length can be developed using FT-Raman system, it would have the same applicability as that shown in this paper and can take advantage of less chance of fluorescence.

In the future, the applicability of the proposed WAI scheme will be evaluated for samples consisting of more complex components such as petroleum products like naphtha. Additionally the analysis of wafer etchant solutions, which are mainly composed of strong inorganic acids and hydrogen peroxide, will also be candidates since Raman spectroscopy is expected to be more sensitive for the analysis of these components compared to infrared and NIR spectroscopy.

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